NPA – Exercise 5

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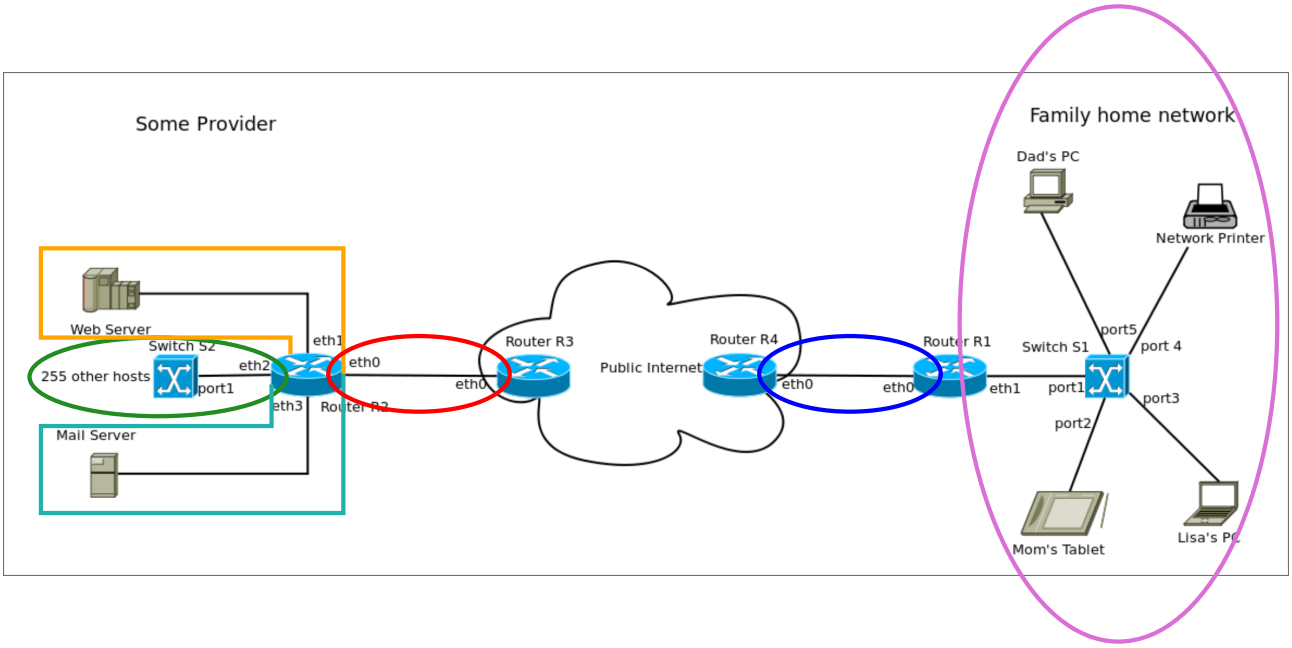
Exercise 1

(a)



(b)

The IP Count is Interfaces + 2. One for the Broadcast and the network address.

Subnet	Number of Interfaces	Number of IPs
Family Home network	5	7
Some Provider Web	2	4
Some Provider Switch	256	258
Some Provider Mail	2	4
Some Provider Uplink	2	4
Family Home Uplink	2	4

Table 1: Summary of Subnets

Exercise 2

(a)

	Network	**Example**
Linux PCs	192.168.0.0/26	192.168.0.42/26
Linux Server	192.168.0.64/26	192.168.0.84/26
Windows PCs A	192.168.1.0/24	192.168.1.42/24
Windows PCs B	192.168.2.0/24	192.168.2.42/24

(b)

		Adress
Linux PCs	Net:	2001:db8:1000::/64
	Example:	2001:db8:1000::c080:2a/64
	Gateway:	2001:db8:1000::c080:1/64
Linux Server	Net:	2001:db8:1000:1::/64
	Example:	2001:db8:1000:1::c080:54/64
	Gateway:	2001:db8:1000:1::c080:41/64
Windows PCs A	Net:	2001:db8:1000:2::/64
	Example:	2001:db8:1000:2::c080:12a/64
	Gateway:	2001:db8:1000:2::c080:101/64
Windows PCs B	Net:	2001:db8:1000:3::/64
	Example:	2001:db8:1000:3::c080:22a/64
	Gateway:	2001:db8:1000:3::c080:201/64

(c)

(i) The NAT router changes the source IP to the single public IPv4 address (its own address) and changes the source port to a yet unused number by the NAT table and stores it.

(ii) The NAT router must keep the LAN side IP address and port (source IP and port of the original packet) and the WAN side IP address and port (new source IP and port of the new packet) when its sent out.

(d)

WAN addr	**WAN port**	**LAN addr**	**LAN port**
133.178.23.14	8000	192.168.1.2	8000
133.178.23.14	8001	192.168.1.3	8000
133.178.23.14	8002	192.168.1.4	8000
133.178.23.14	80	192.168.0.66	80

Table 2: NAT table

(e)

If an IP packet was sent from 141.42.4.130 to winA002 with internal IP 192.168.1.3, the NAT router would change the destination address (and port if necessary) of the packet from its own external address and port, to the internal IP of winA002 and the respective port of the desired application.

Its own external address and port together with the internal address and port of winA002 are saved to the NAT table.

(f)

(i) 64:ff9b::cb00:7128 (64:ff9b::203.0.113.40)

(ii) a) IPv6 link

Source IP	Source port	Dest. IP	Dest. port	Protocol
2001:db8:1000:2::c080:12a	5000	64:ff9b::cb00:7128	443	TCP

b) IPv4 link

Source IP	Source port	Dest. IP	Dest. port	Protocol
133.178.23.14	5000	203.0.113.40	443	TCP

(iii) The NAT64 translates the source IP address and destination IP address. The source port can be changed when necessary, but does not need to be translated in terms of IPv4/IPv6 translation.